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IFMBE

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Optimization of Electrode Configuration for Bladder-Targeted Temporal Interference Stimulation Using a Three-Dimensional Rat Model

Presenter | Thar Htet Nyan

Principal Investigator | Eunkyoung Park

Department of Biomedical Engineering, Soonchunhyang University, Asan, Republic of Korea

01

Background & Research Objectives

AGENDA

01 Background & Research Objectives

02 Phantom Experimental Validation

03 2D Rat Models: Preliminary Placement

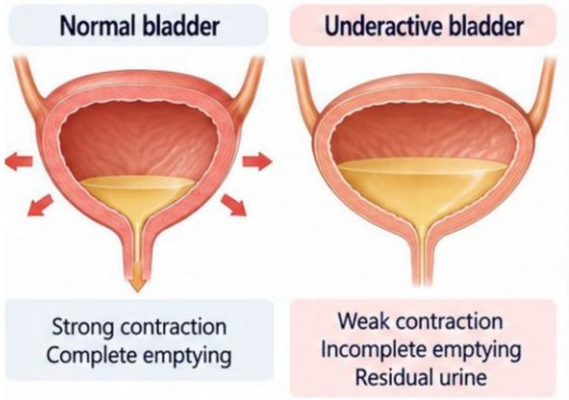
04 3D Rat Models: Optimization

05 Electrode Geometry Comparison

06 Limitations and Future Work

Background

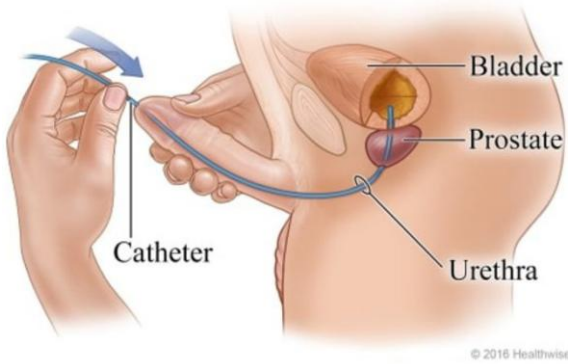
Underactive Bladder (UAB)



- Normal urination requires coordinated detrusor muscle contraction and urethral relaxation.
- In UAB, impaired detrusor contraction can lead to incomplete bladder emptying and urinary retention.

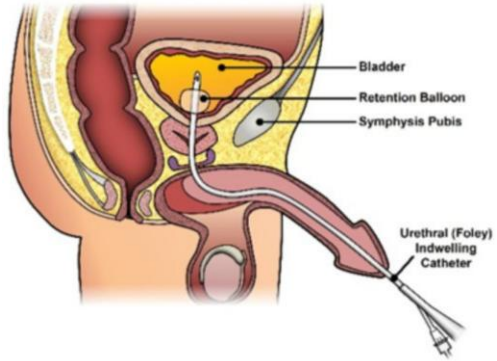
Current Treatment

Intermittent Catheterization



- Requires repeated catheter insertion for bladder emptying.
- May cause discomfort, urethral trauma, and urinary tract infections.

Indwelling Catheterization



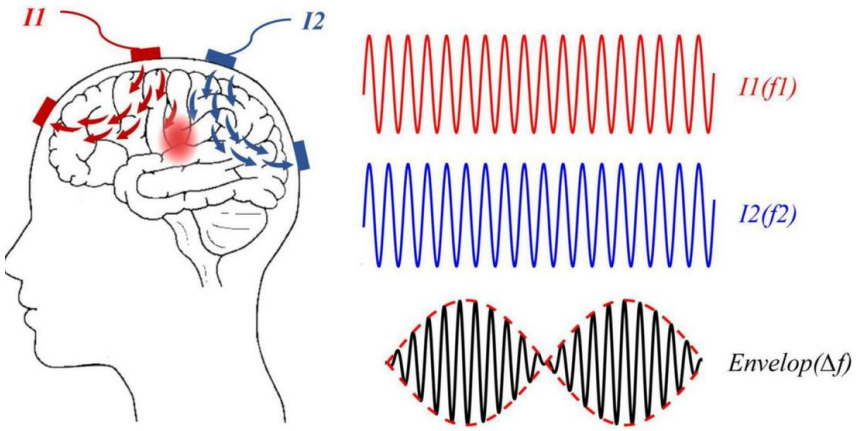
- Requires continuous catheter placement and maintenance.
- Prolonged use increases the risk of infection and catheter-related complications.

Research Need **A non-invasive neuromodulation approach capable of selectively targeting bladder-related structures warrants investigation.**

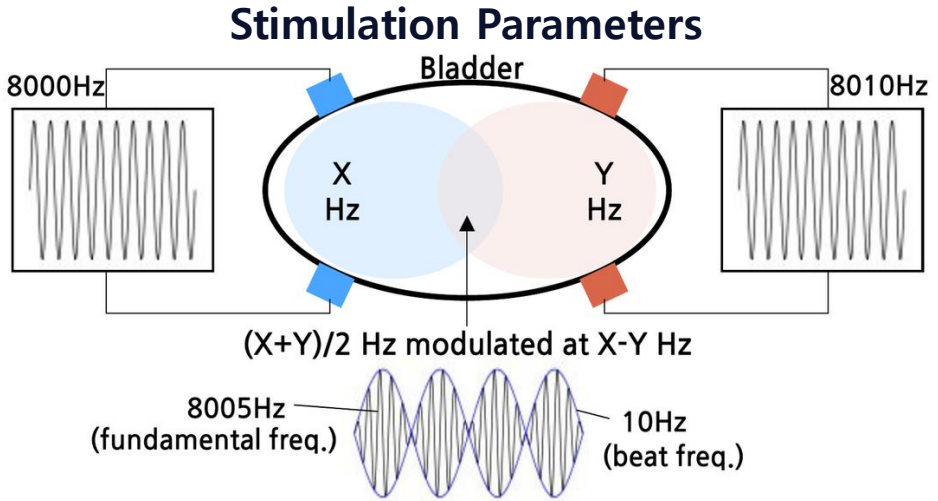
Temporal Interference Stimulation

Principle of Temporal Interference Stimulation (TIS)

- Temporal Interference Stimulation (TIS) is a non-invasive electrical neuromodulation technique designed to target deep neural structures.
- It uses two slightly different high-frequency electric fields to generate a low-frequency amplitude-modulation envelope.
- Primarily investigated for deep brain stimulation, TIS is also being explored for spinal cord and peripheral neuromodulation applications.



Deep Brain Stimulation



1. Guo W, He Y, Zhang W, Sun Y, Wang J, Liu S and Ming D (2023) A novel non-invasive brain stimulation technique: "Temporally interfering electrical stimulation". Front. Neurosci. 17:1092539. doi: 10.3389/fnins.2023.1092539

Research Objectives & Workflow

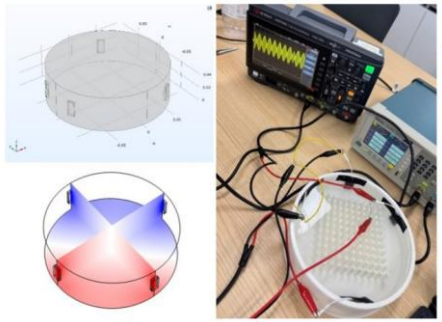
Main Objective

- To optimize electrode placement for bladder-targeted temporal interference stimulation using experimental validation and computational modeling.

Overall Workflow

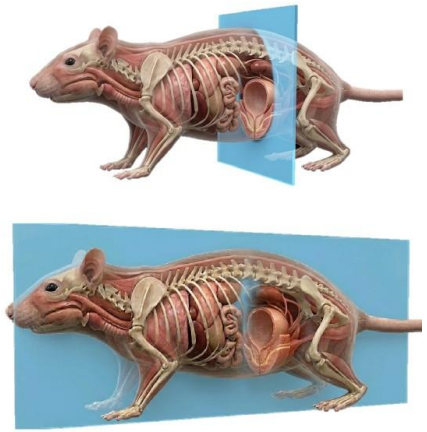
I. Phantom

Experimental validation



II. 2D Rat Model

Preliminary placement



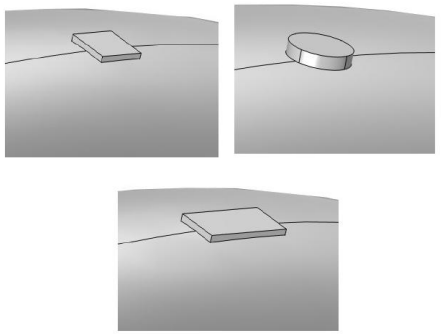
III. 3D Rat Model

3D optimization



IV. Geometry

Electrode shape comparison



02

Phantom Experimental Validation

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02 **Phantom Experimental Validation**

03 2D Rat Models: Preliminary Placement

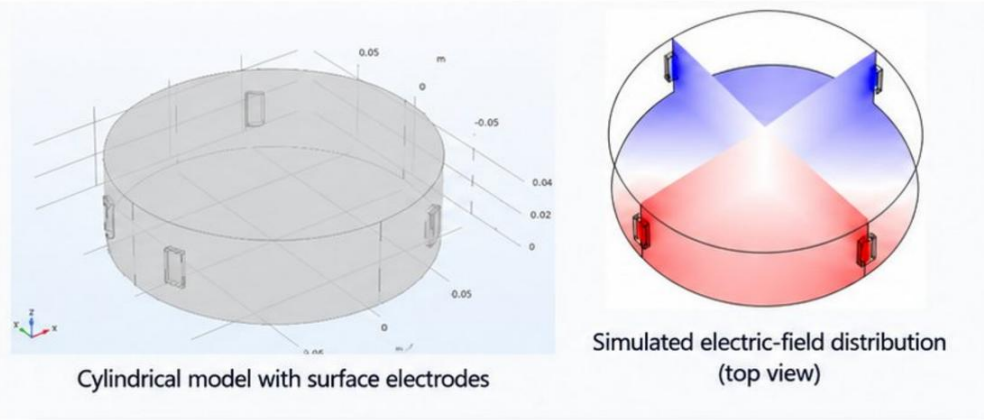
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06 Limitations and Future Work

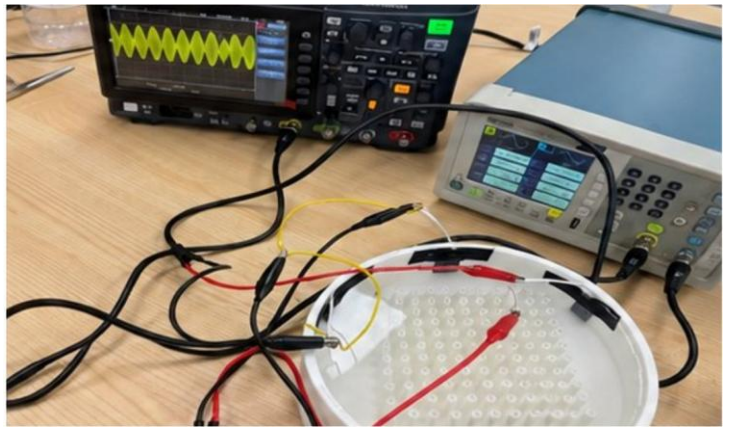
Phantom Experimental Validation

COMSOL Stimulation



- COMSOL Multiphysics 6.3
- Cylinder model: R = 95 mm, H = 50 mm
- Saline conductivity: 1.7 S/m; gel: 1.0 S/m
- Air treated as an insulator

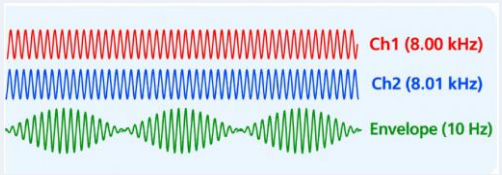
Physical Saline Phantom



- 3D-printed PLA circular container
- 0.9% NaCl saline, σ ≈ 1.7 S/m
- Surface electrodes: 1 × 2 cm
- 11 × 11 measurement grid (10 mm spacing)
- Six electrode configurations tested

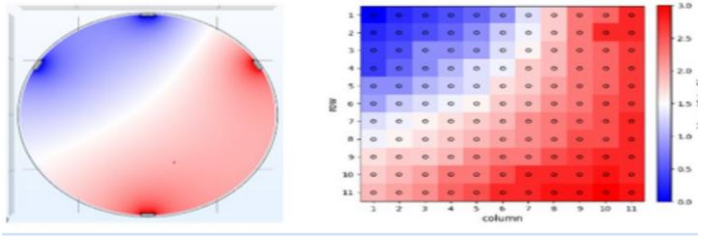
Stimulation Parameter

Channel 1	Channel 2	Frequency Difference (Envelope)
8.00 kHz (3.1 Vpp)	8.01 kHz (3.1 Vpp)	10 Hz

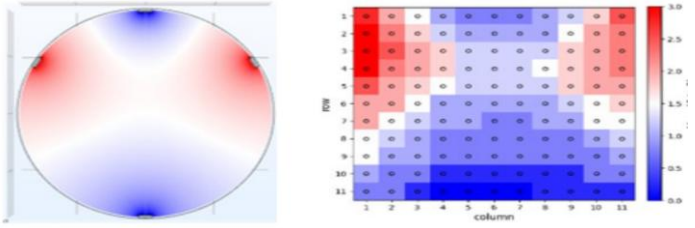


Phantom Experimental Validation: Result

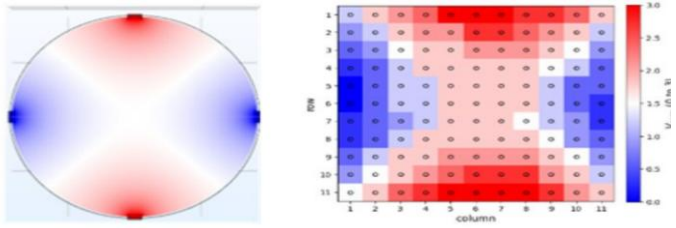
Case 1 COMSOL (Simulation) Experiment (Phantom)



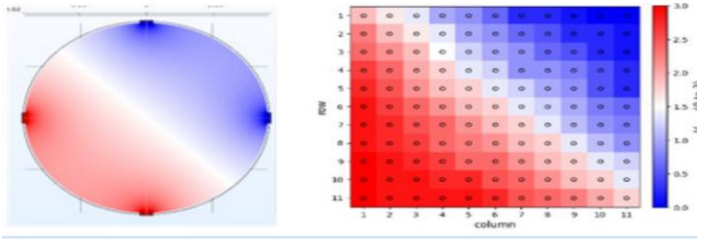
Case 2 COMSOL (Simulation) Experiment (Phantom)



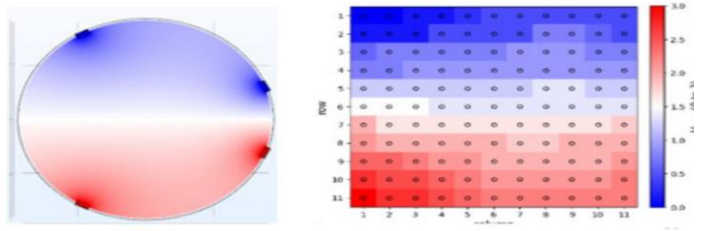
Case 3 COMSOL (Simulation) Experiment (Phantom)



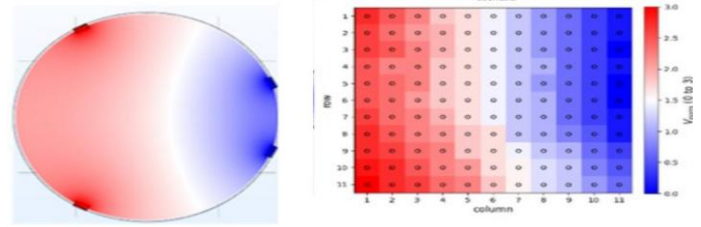
Case 4 COMSOL (Simulation) Experiment (Phantom)



Case 5 COMSOL (Simulation) Experiment (Phantom)



Case 6 COMSOL (Simulation) Experiment (Phantom)



Experimental measurements showed comparable spatial field-distribution trends to the COMSOL simulations.

03

2D Rat Models Preliminary Placement

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01 Background & Research Objectives

02 Phantom Experimental Validation

03 **2D Rat Models: Preliminary Placement**

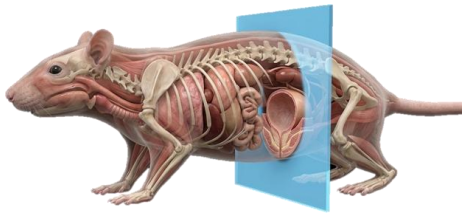
04 3D Rat Models: Optimization

05 Electrode Geometry Comparison

06 Limitations and Future Work

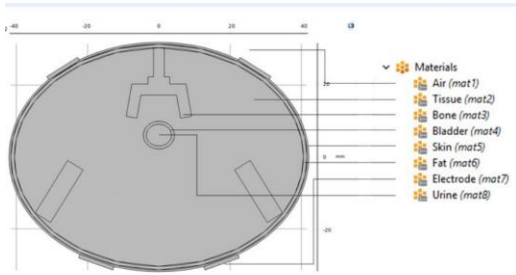
2D Rat Models: Stimulation Setup

Transverse Model (Cross-sectional)



40 × 30 mm
2D model size

COMSOL Geometry



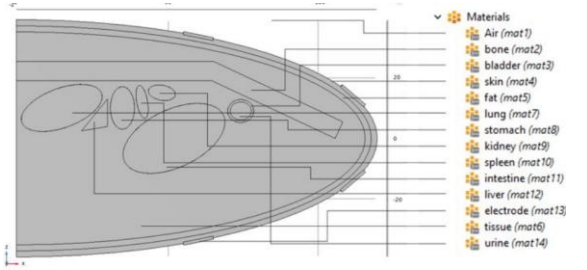
- Bladder-centered cross-section
- Model size: 40 × 30 mm
- Surface electrode placement
- 2D electric-field analysis

Sagittal Model (Longitudinal)



230 × 70 mm
2D model size

COMSOL Geometry



- Bladder-containing longitudinal section
- Model size: 230 × 70 mm
- Surface electrode placement
- 2D electric-field analysis

Material Properties (Electrical Conductivity)

Transverse model

- Tissue: σ = 0.339 S/m
- Bone: σ = 0.0204 S/m
- Bladder: σ = 0.2125 S/m
- Skin: σ = 0.00182 S/m
- Fat: σ = 0.0429 S/m
- Urine: σ = 1.5 S/m
- Electrode gel: σ = 1.0 S/m
- Air: σ = 1×10^{-14} S/m (insulator)

Sagittal model

Same as transverse model, plus:

- Lung: σ = 0.0919 S/m
- Stomach: σ = 0.529 S/m
- Kidney: σ = 0.339 S/m
- Spleen: σ = 0.1100 S/m
- Intestine: σ = 0.557 S/m
- Liver: σ = 0.0514 S/m

Stimulation Parameters

f_1 = 8,000 Hz

f_2 = 8,010 Hz

Current = 3 mA

Frequency difference (envelope) = 10 Hz

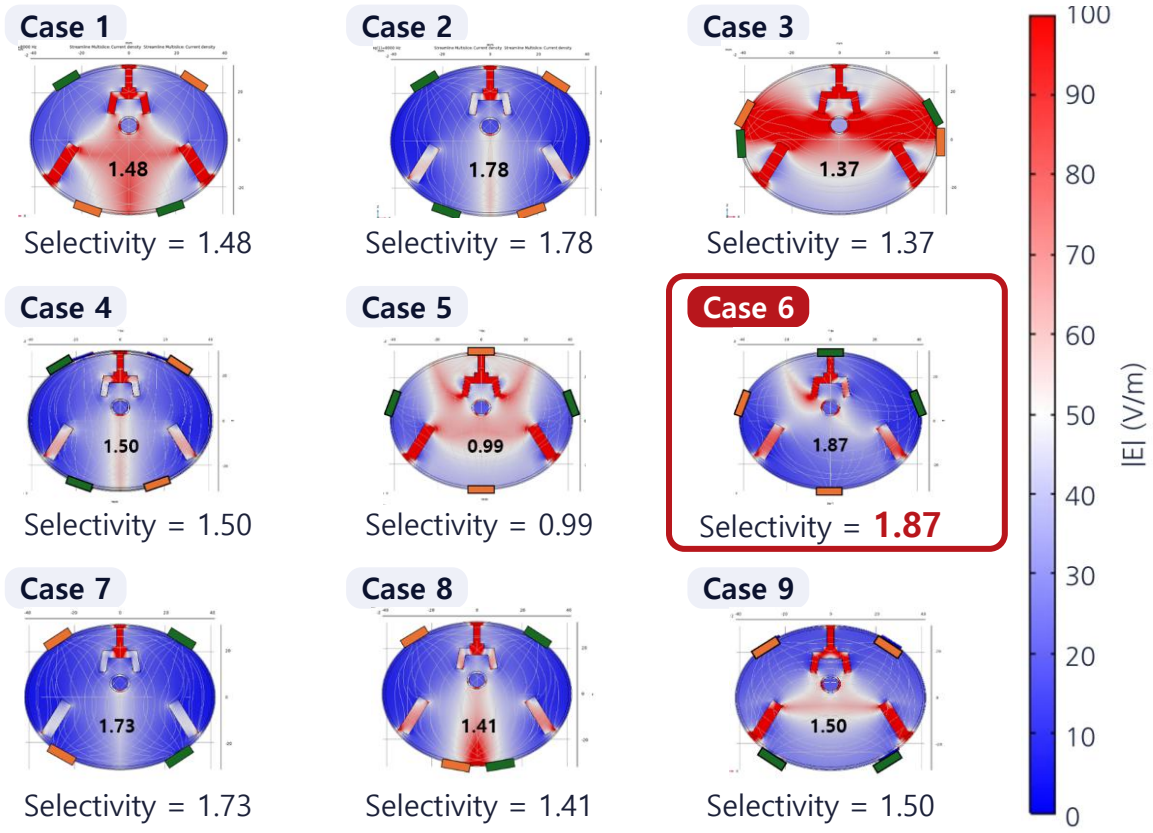
Evaluation Metrics

Selectivity: $\frac{E_{bladder}}{E_{tissue}}$

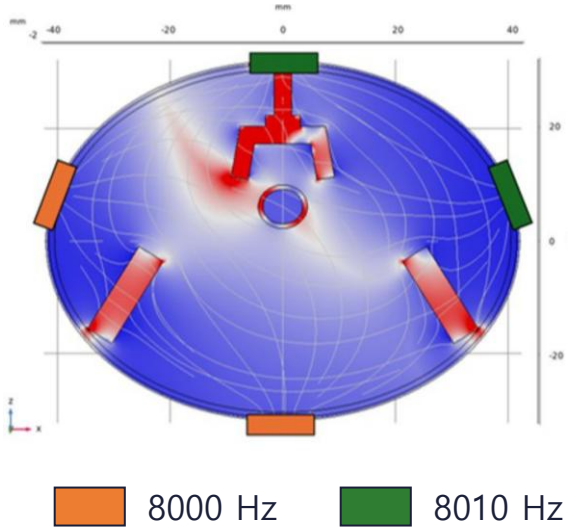
- Electric-field magnitude (V/m)
- Temporal interference (TI) envelope

2D Transverse Model: Electrode Placement Optimization

Nine Electrode Configurations (Transverse Section)



Optimal Configuration: Case 6

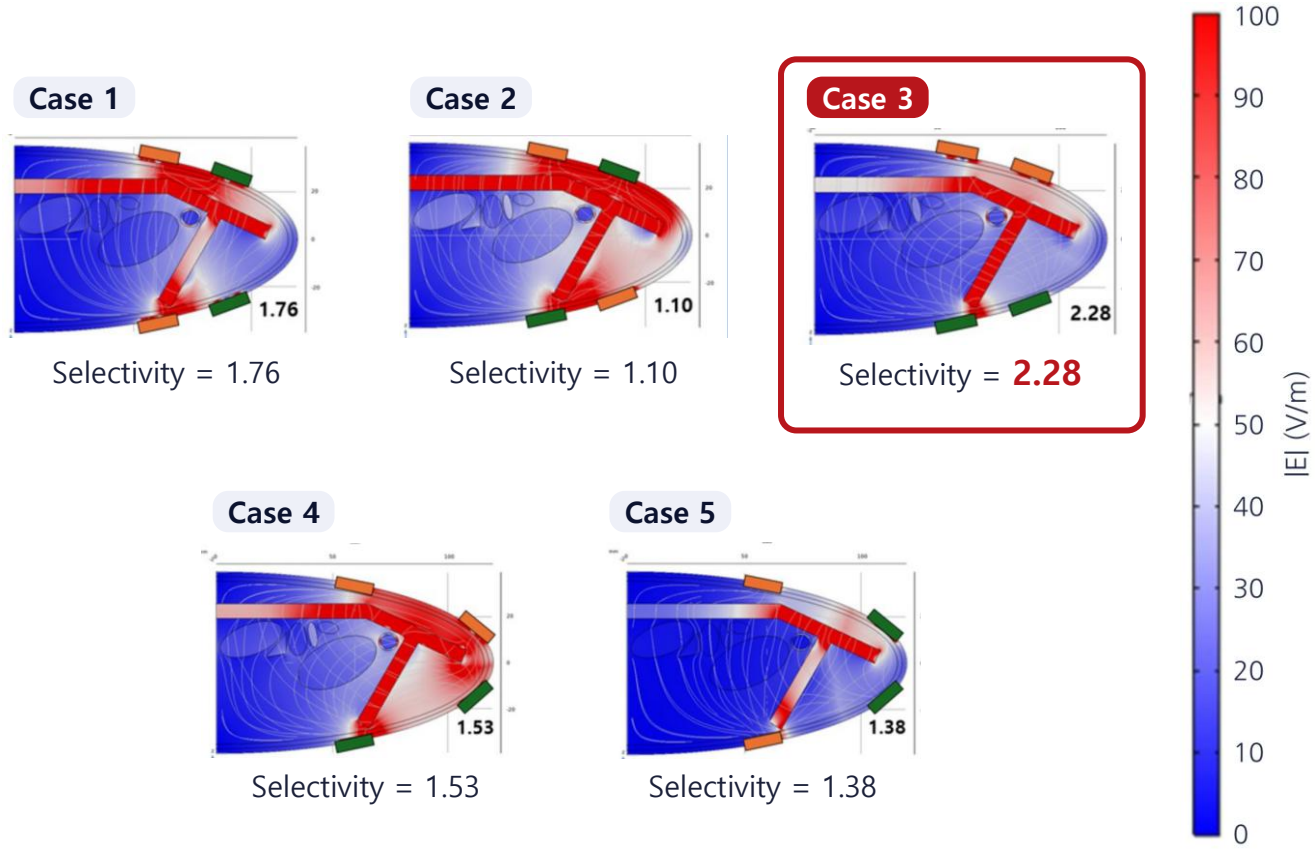


Mean bladder field (V/m)	**40.11**
Mean tissue field (V/m)	**21.40**
Bladder-to-tissue selectivity	**1.87**
Electrode arrangement	**Top–Diagonal**

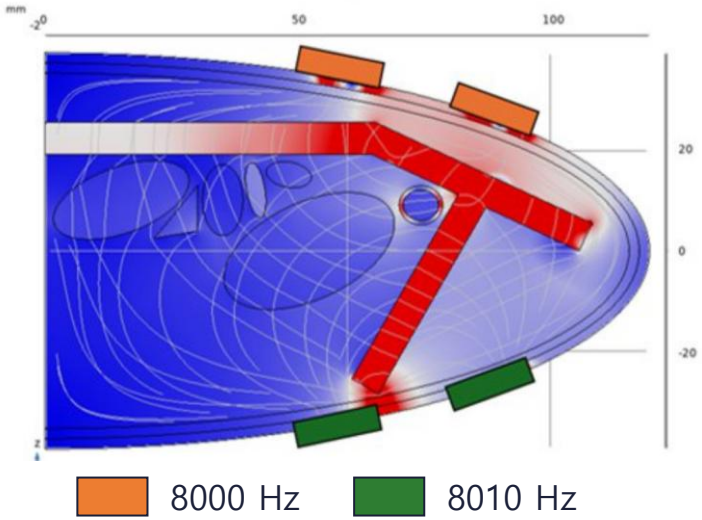
Case 6 achieved the highest bladder-to-tissue selectivity among nine transverse electrode configurations.

2D Sagittal Model: Electrode Placement Optimization

Five Electrode Configurations (Sagittal Section)



Optimal Configuration: Case 3



Mean bladder field (V/m)	**42.85**
Mean tissue field (V/m)	**18.79**
Bladder-to-tissue selectivity	**2.28**
Electrode arrangement	**Top–Bottom Diagonal**

Case 3 achieved the highest bladder-to-tissue selectivity among five sagittal electrode configurations.

04

3D Rat Models Optimization

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01 Background & Research Objectives

02 Phantom Experimental Validation

03 2D Rat Models: Preliminary Placement

04 **3D Rat Models: Optimization**

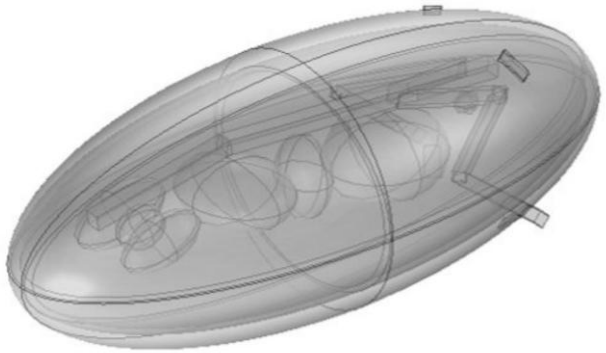
05 Electrode Geometry Comparison

06 Limitations and Future Work

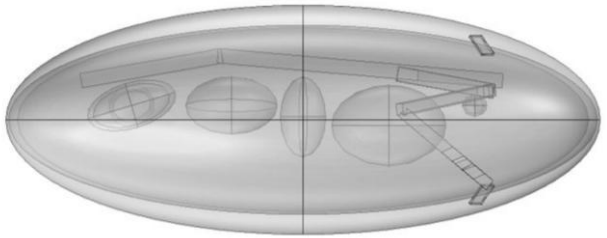
3D Rat Model: Simulation Setup

3D Anatomical Model

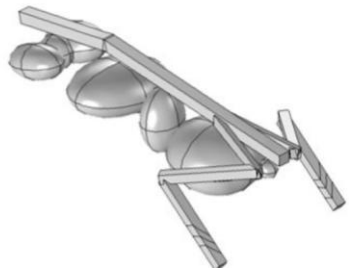
Oblique view



Orthogonal view



Structures



Model Setup and Simulation Parameters

Model Setup

Model dimensions	104 × 44 × 40 mm
Anatomical tissue structures	Included
Analysis	3D electric-field analysis

Stimulation Parameters

Stimulation frequencies	f_1 = 8,000 Hz | f_2 = 8,010 Hz (Δf = 10 Hz)
Stimulation current	3 mA
Electrode configurations	5 configurations (surface electrodes)

Evaluation Metrics

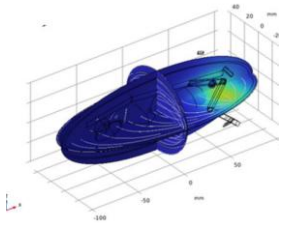
Selectivity	$S_{B/T} = \frac{\bar{E}_{bladder}}{\bar{E}_{tissue}}$
Electric-field magnitude	(V/m)
TI envelope	Temporal interference (TI) envelope

Objective: | Optimize electrode placement to maximize the electric field in the bladder while minimizing off-target tissue exposure.

3D Electrode Configuration Comparison

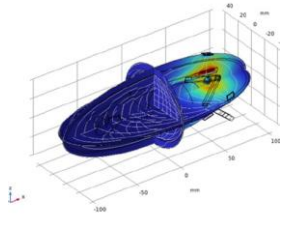
Five Surface-Electrode Configurations (3D Model)

Case 1



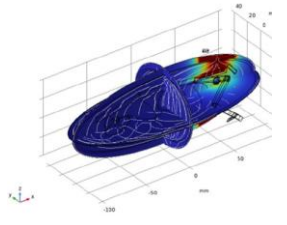
Selectivity = 6.83

Case 2



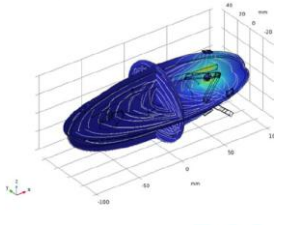
Selectivity = 5.47

Case 3



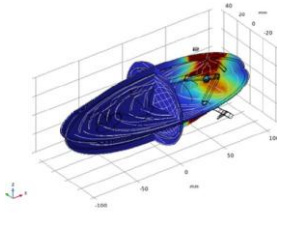
Selectivity = 6.20

Case 4

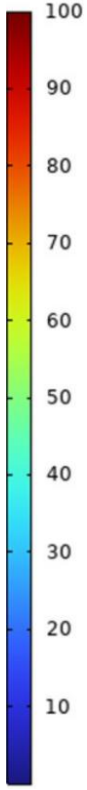


Selectivity = **7.96**

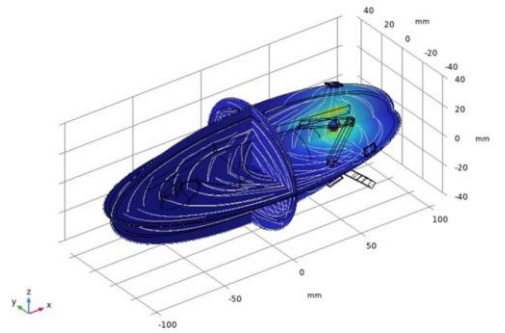
Case 5

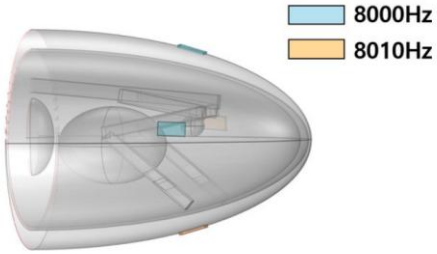


Selectivity = 5.12

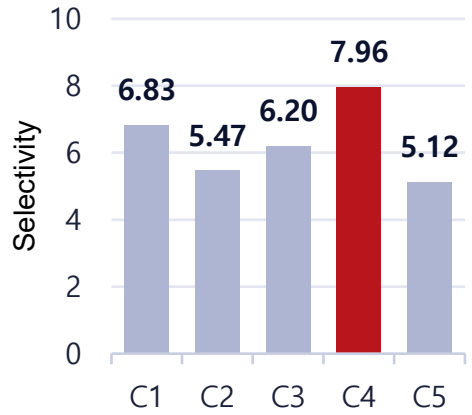


Optimal Configuration: Case 4





Bladder/Tissue Selectivity



Quantitative Results

Mean bladder field	**63.30 V/m**
Mean surrounding-tissue field	**7.95 V/m**
Bladder-to-tissue selectivity	**7.96**

Case 4 achieved the highest bladder-to-tissue selectivity among the five 3D electrode configurations.

Optimal Electrode Placement Strategy

Stimulation Channels

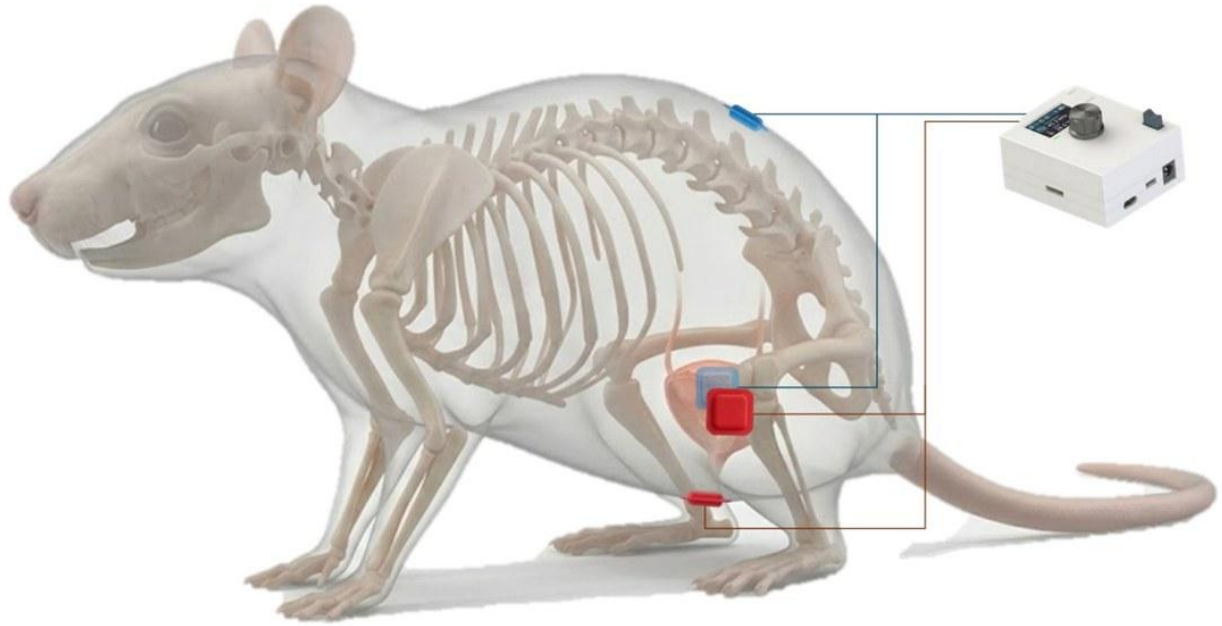
Channel 1 (Top-Side Pair)

Electrode	**Upper (above bladder)**
Return	**Lateral (side)**

Channel 2 (Bottom-Side Pair)

Electrode	**Lower (below bladder)**
Return	**Opposite lateral side**

Optimal Electrode Placement



Channel 1 - Top-Side Pair Channel 2 - Bottom-Side Pair

Two electrode pairs: Channel 1 (Top-Side) uses an upper electrode; Channel 2 (Bottom-Side) uses a lower one.

05

Electrode Geometry Comparison

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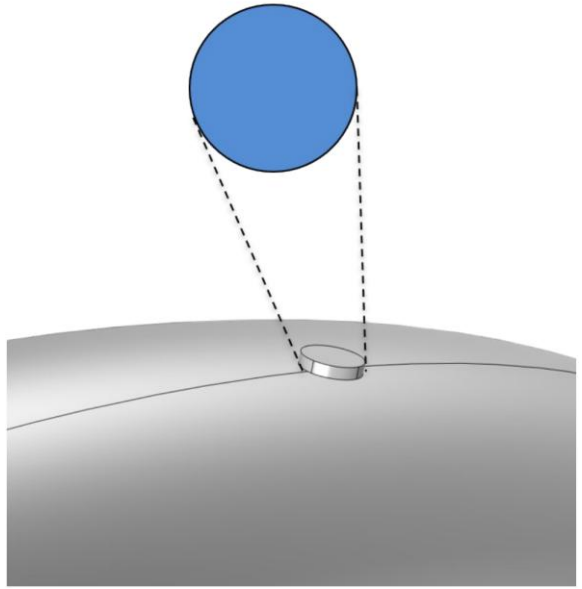
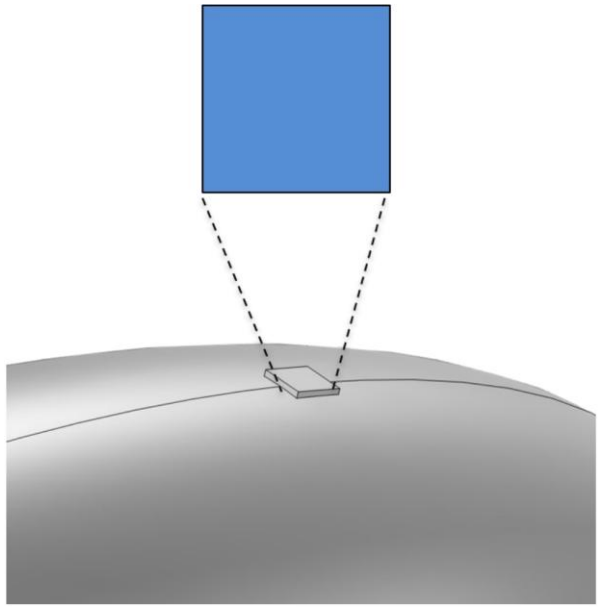
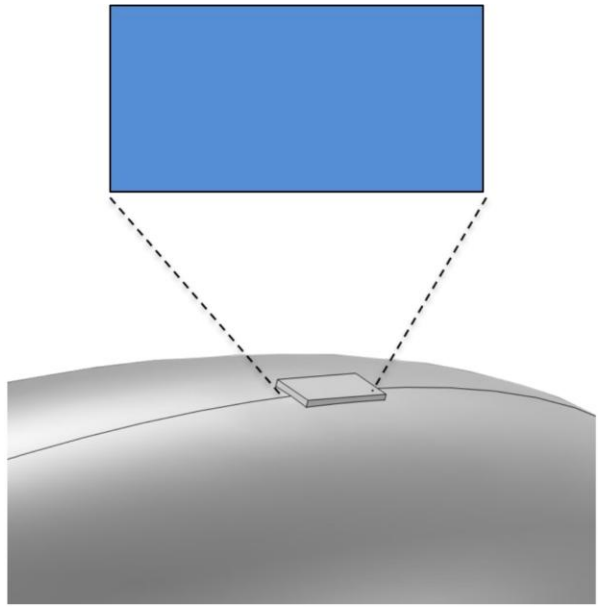
03 2D Rat Models: Preliminary Placement

04 3D Rat Models: Optimization

05 **Electrode Geometry Comparison**

06 Limitations and Future Work

Electrode Geometry Comparison

Circle	Square	Rectangle
Radius: 2.5 mm	Length × Width: 3 × 3 mm	Length × Width: 3 × 6 mm
		
Selectivity **6.824**	Selectivity **6.830**	Selectivity **6.828**

Only minor differences among the three geometries; the 3 × 3 mm square gave the highest selectivity (6.830).

06

Limitations and Future Work

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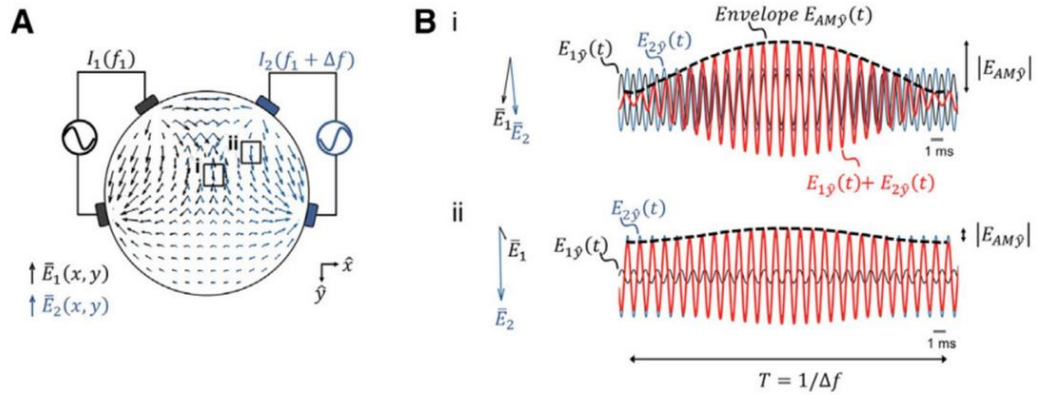
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Limitations

Direction-Dependent TI Modulation

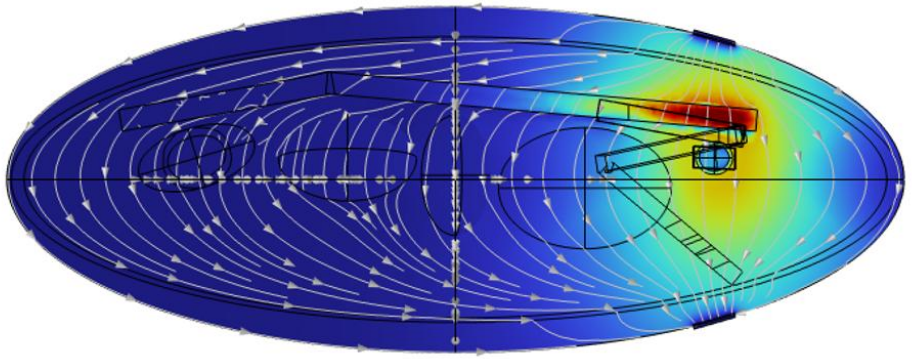


$$\Delta E(\hat{n}) = 2\min\left(|\mathbf{E}_1 \cdot \hat{n}|, |\mathbf{E}_2 \cdot \hat{n}|\right)$$

TI modulation amplitude depends on both electric-field magnitude and direction.

- The TI envelope amplitude varies with the projection of the two electric fields along a specific direction.
- Even at the same location, different field orientations can produce different modulation amplitudes.

Limitations of Magnitude-Based TI Analysis



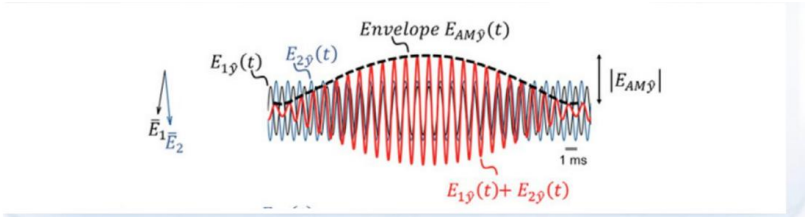
Electric-Field Distribution in the Current Electrode Configuration

- Electrode placement influences both the magnitude and orientation of electric fields within the tissue.
- The present study used a magnitude-based TI envelope approximation without explicitly accounting for directional dependence.
- Consequently, magnitude-based analysis may not fully represent the effective TI modulation along specific directions.

2. Grossman N, et al. Noninvasive Deep Brain Stimulation via Temporally Interfering Electric Fields. Cell. 2017;169(6):1029–1041.e16. doi:10.1016/j.cell.2017.05.024.

Future Work

Directional Dependent Study



In Vivo Experiment



Clinical Study



1. **Perform direction-dependent TI envelope analysis to further optimize electrode placement and stimulation selectivity.**

2. **Conduct *in vivo* experiments to evaluate the safety and efficacy of TIS in animal models of UAB.**

3. **Conduct clinical studies to assess the feasibility and therapeutic potential of non-invasive TIS in patients with UAB.**

Conclusion

1. Phantom experiments validated the COMSOL simulation through comparable electric-field distributions.
2. 2D models revealed placement-dependent field behavior and guided electrode placement for 3D modeling.
3. 3D optimization identified Case 4, using a top-to-side arrangement for Channel 1 and a bottom-to-side arrangement for Channel 2, as the most selective configuration for bladder-targeted TIS.
4. Electrode geometries showed only minor differences in selectivity.
5. These findings provide a foundation for future experimental validation and development of bladder-targeted TIS.

Thank You